

$$\{x_1, y_1\} \quad \{x_2, y_2\} \quad \dots \quad \{x_n, y_n\}$$

P_1

P_2

P_n

$$Z = g(x, y)$$

$$z_1 = g(x_1, y_1)$$

$$z_2 = g(x_2, y_2)$$

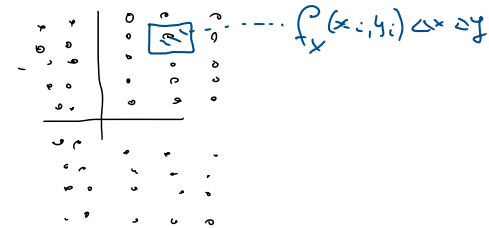
$$z_n = g(x_n, y_n)$$

$$E[Z] = \sum_{i=1}^n g(x_i, y_i) P_i$$

$$g(x_1, y_1) P_1 + \dots + g(x_n, y_n) P_n$$

$$g(x_n, y_n) P_n$$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f_{X,Y}(x, y) dx dy$$



$$= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} g(x, y) f_{X,Y}(x, y) dy \right] dx = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} g(x, y) f_{X,Y}(x, y) dx \right] dy$$

$$Z = aX + bY = g(X, Y) \quad g(x, y) = ax + by$$

$$\int g(x, y) f_{X,Y}(x, y) dy = \int (ax + by) f_{X,Y}(x, y) dy$$

$$= \int ax f_{X,Y}(x, y) dy + \int by f_{X,Y}(x, y) dy$$

$$= a \times \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy + b \int_{-\infty}^{+\infty} y f_{X,Y}(x, y) dy$$

$$\int_{-\infty}^{+\infty} \left[a \times f_X(x) + b \int_{-\infty}^{+\infty} y f_{X,Y}(x, y) dy \right] dx$$

$$= a \int_{-\infty}^{+\infty} x f_X(x) dx + b \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} y f_{X,Y}(x, y) dy \right] dx = a E[X] + b E[Y]$$

$$\int_{-\infty}^{+\infty} y \left[\int_{-\infty}^{+\infty} f_{X,Y}(x, y) dx \right] dy$$

$f_Y(y)$

$E[Y] + b E[Y]$

$$\underbrace{\int_{-\infty}^{+\infty} y f_Y(y) dy}_{E[Y]}$$

$$E[aX + bY] = aE[X] + bE[Y]$$

$$W = \underbrace{aX + bY + cZ}_{Q} = 1 \cdot Q + cZ$$

$$E[W] = 1 \cdot E[Q] + c E[Z] = a E[X] + b E[Y] + c E[Z]$$

$$Z = \sum_{i=1}^m a_i X_i = 1 \cdot Q + a_m X_m$$

$$Q = \sum_{i=1}^{m-1} a_i X_i$$

$$E[Q] = \sum_{i=1}^{m-1} a_i E[X_i]$$

$$E[Z] = E[Q] + a_m E[X_m]$$

$$= \left\{ \sum_{i=1}^{m-1} a_i E[X_i] \right\} + a_m E[X_m] = \sum_{i=1}^m a_i E[X_i]$$

$$f_{X,Y}(x,y) = f_Y(y) f_{X|Y}(x|y)$$

$$E[g(X,Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x,y) f_{X,Y}(x,y) dx dy = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x,y) f_Y(y) f_{X|Y}(x|y) dx dy$$

$$= \int_{-\infty}^{+\infty} f_Y(y) \left[\int_{-\infty}^{+\infty} g(x,y) f_{X|Y}(x|y) dx \right] dy$$

$$X|_{Y=y}$$

$$h(y) = E[Z|_{Y=y}]$$

$$Z|_{Y=y} = g(X,Y)|_{Y=y} = g(X,y)$$

$$= \int_{-\infty}^{+\infty} h(y) f_Y(y) dy = E[h(Y)]$$

$$g(x,y) = (x - \mu_x)(y - \mu_y)$$

$$\text{Cov}[X,Y] = E[(X - \mu_x)(Y - \mu_y)] = E[g(X,Y)]$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x,y) f_{X,Y}(x,y) dx dy = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - \mu_x)(y - \mu_y) f_{X,Y}(x,y) dx dy$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - \mu_x)(y - \mu_y) f_X(x) f_Y(y) dx dy$$

Indip

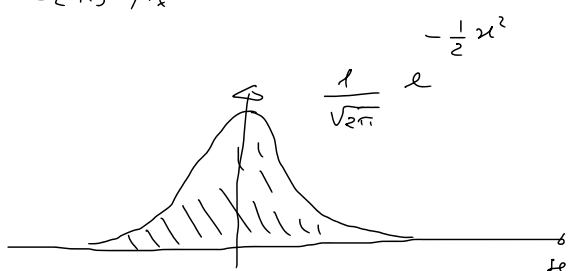
$$= \int_{-\infty}^{+\infty} (y - \mu_y) f_y(y) \left[\int_{-\infty}^{+\infty} (x - \mu_x) f_x(x) dx \right] dy$$

$$= \int_{-\infty}^{+\infty} (x - \mu_x) f_x(x) dx \int_{-\infty}^{+\infty} (y - \mu_y) f_y(y) dy = 0$$

$$\int_{-\infty}^{+\infty} x f_x(x) dx - \int_{-\infty}^{+\infty} \mu_x f_x(x) dx = \mu_x - \mu_x \int_{-\infty}^{+\infty} f_x(x) dx = 0$$

$\int_{-\infty}^{+\infty} f_x(x) dx = 1$

$E[X] = \mu_x$

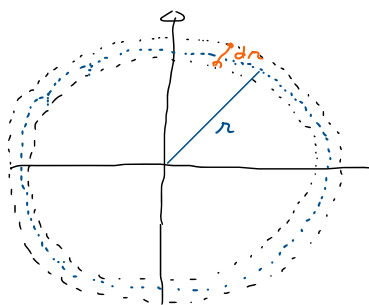


$$\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = 1$$

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_x(x) dx$$

$$\left[\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \right]^2 = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$



$$\int_0^{+\infty} \left[\frac{1}{2\pi} e^{-\frac{1}{2}r^2} \right] (2\pi r dr) = \int_0^{+\infty} r e^{-\frac{1}{2}r^2} dr = \left[-e^{-\frac{1}{2}r^2} \right]_{r=0}^{r=+\infty} = 1$$

$$= \int_{-\infty}^{+\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x \left[x e^{-\frac{1}{2}x^2} \right] dx = \frac{1}{\sqrt{2\pi}} \left\{ \left[x \left(-e^{-\frac{1}{2}x^2} \right) \right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} 1 \cdot \left(-e^{-\frac{1}{2}x^2} \right) dx \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}x^2} dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = 1$$

$$Y = aX \quad f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y}{a}\right)$$

$$X = \sigma_X X_0 + \mu_X$$

$$Y = \sigma_X X_0 \quad f_Y(y) = \frac{1}{\sigma_X} f_{X_0}\left(\frac{y}{\sigma_X}\right) = \frac{1}{\sigma_X} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{y}{\sigma_X}\right)^2} = \frac{1}{\sqrt{2\pi} \sigma_X} e^{-\frac{y^2}{2\sigma_X^2}}$$

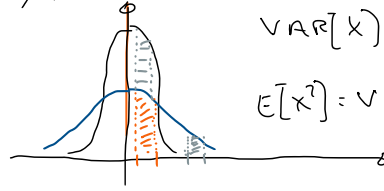
$$E[Y] = \sigma_X E[X_0] = 0$$

$$E[X] = E[\sigma_X X_0] + \mu_X = \mu_X$$

$$\text{VAR}[X] = E[(X - \mu_X)^2] = E[Y^2] = \sigma_X^2$$

$$E[Y^2] = \sigma_X^2 = \sigma_X^2 E[X_0^2] = \sigma_X^2$$

$$E[X^2] = \text{VAR}[X] + \mu_X^2 = \sigma_X^2 + \mu_X^2$$



$$f_X(x) = f_{Y+\mu_X}(x) = f_Y(x - \mu_X) = \frac{1}{\sqrt{2\pi} \sigma_X} e^{-\frac{1}{2} \frac{(x - \mu_X)^2}{\sigma_X^2}}$$

$$F_X(x) = P[X \leq x] = 1 - P[X > x]$$

C D F
complementary

$$P[X \in (a, b)] = F_X(b) - F_X(a) = Q_X(a) - Q_X(b)$$

$$Q(x) = P[X_0 > x] = \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2} dx$$

$$\begin{aligned} P_{X_0}[x; \Delta x] &= P\left[x - \frac{\Delta x}{2} < X_0 < x + \frac{\Delta x}{2}\right] = F_{X_0}\left(x + \frac{\Delta x}{2}\right) - F_{X_0}\left(x - \frac{\Delta x}{2}\right) \\ &= \left[1 - Q\left(x + \frac{\Delta x}{2}\right)\right] - \left[1 - Q\left(x - \frac{\Delta x}{2}\right)\right] \\ &= Q\left(x - \frac{\Delta x}{2}\right) - Q\left(x + \frac{\Delta x}{2}\right) \end{aligned}$$

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2)$$

$$X_0 = Y = \frac{X - \mu_X}{\sigma_X} = g(X)$$

$$g(x) = \frac{x - \mu_X}{\sigma_X}$$

$$g(x) = y \Leftrightarrow \frac{x - \mu_X}{\sigma_X} = y \Leftrightarrow x = \sigma_X y + \mu_X = x_1(y)$$

$$x_1(y) = \sigma_X y + \mu_X$$

$$f_Y(y) = \sigma_X f_X(x_1(y)) = \sigma_X f_X[\sigma_X y + \mu_X] = \frac{1}{\sqrt{2\pi} \sigma_X} e^{-\frac{1}{2} \frac{(\sigma_X y + \mu_X - \mu_X)^2}{\sigma_X^2}}$$

$$x_1'(y) = \sigma_X$$

$$|x_1'(y)| = \sigma_X$$

$$= \sigma_X \frac{1}{\sqrt{2\pi} \sigma_X} e^{-\frac{1}{2} y^2}$$

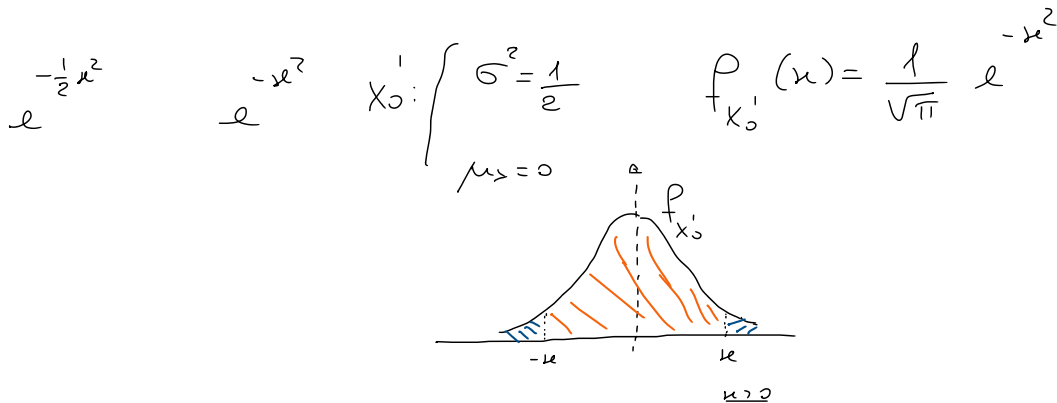
$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} y^2}$$

$$X \geq x \Leftrightarrow \sigma_X X_0 + \mu_X \geq x \Leftrightarrow X_0 \geq \frac{x - \mu_X}{\sigma_X} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} y^2} = f_{X_0}(y)$$

$$X_0 = \frac{X - \mu_x}{\sigma_x} \Leftrightarrow X = \sigma_x X_0 + \mu_x$$

$$P[X > u] = P\left[X_0 \geq \frac{u - \mu_x}{\sigma_x}\right] = Q\left(\frac{u - \mu_x}{\sigma_x}\right)$$

$$P[N(\mu; \sigma^2) \geq u] = Q\left(\frac{u - \mu_x}{\sigma_x}\right)$$



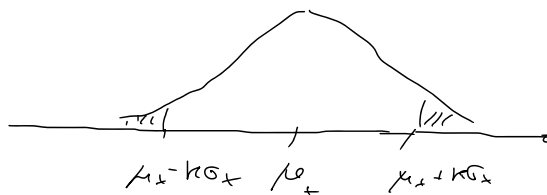
erf
erfc

$$\text{erfc}(u) = 2 \int_u^{+\infty} f_{X_0'}(t) dt = \frac{2}{\sqrt{\pi}} \int_u^{+\infty} e^{-t^2} dt$$

$$\text{erfc}(u) = \frac{2}{\sqrt{\pi}} \int_u^{+\infty} e^{-t^2} dt \xrightarrow{\frac{\beta}{\sqrt{2}} = t} = \frac{2}{\sqrt{\pi}} \int_{u\sqrt{2}}^{+\infty} e^{-\frac{1}{2}\beta^2} \frac{d\beta}{\sqrt{2}} = 2 \int_{u\sqrt{2}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\beta^2} d\beta = 2P[X_0 \geq u\sqrt{2}] = 2Q(u\sqrt{2})$$

$f_{X_0'}(\beta)$

$$Q(u) = \frac{1}{2} \text{erfc}\left(\frac{u}{\sqrt{2}}\right)$$



$$X \sim N(\mu_x, \sigma_x^2)$$

$$\mu_x - k\sigma_x < X < \mu_x + k\sigma_x$$

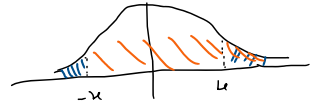
$$-k\sigma_x < X - \mu_x < k\sigma_x$$

$$-k < \underbrace{\frac{X - \mu_x}{\sigma_x}}_{X_0} < k \Rightarrow |X_0| < k$$



$$P[|X_0| < k] = 1 - 2 P[X_0 > k]$$

$$P[X \notin (\mu_x - k\sigma_x, \mu_x + k\sigma_x)] = 2 P[X_0 > k] = 2 Q(k)$$



$$Q(-u) = P[X > -u] = 1 - P[X < -u] = 1 - P[X > u] = 1 - Q(u)$$

$$\frac{d}{du} \int_u^{+\infty} e^{-\frac{1}{2}d^2} dd = \frac{d}{du} \left[- \int_{+\infty}^u e^{-\frac{1}{2}d^2} dd \right] = - e^{-\frac{1}{2}u^2}$$

$$P[i] = \begin{cases} \frac{1}{2} & i=6 \\ \frac{1}{10} & i \in \{1, \dots, 5\} \end{cases}$$

$$u = \frac{1}{2}$$

$$P[X = u] \\ \frac{k=4}$$

D_0 D_0

X_1

X_2

$$P[(X_1, X_2) = (6, 6)]$$

$$P_{X_1/D_0}[i] = \begin{cases} \frac{1}{6} & i \in \{1, \dots, 6\} \end{cases}$$

$$P[(X_1, X_2) = (6, 5)]$$

$$P[(X_1, X_2) = (5, 6)]$$

$$P_{X_2/D_0}[i] = \begin{cases} \frac{1}{10} & i=6 \\ \frac{1}{20} & i \in \{1, \dots, 5\} \end{cases}$$

$$P[(X_1 = \cancel{6}) \text{ AND } (X_2 = \cancel{6})] = P[X_1 = \cancel{6}, X_2 = \cancel{6}] =$$

$$= P(e_0) P[X_1 = \cancel{6}, X_2 = \cancel{6} | e_0] + P(\bar{e}_0) P[X_1 = \cancel{6}, X_2 = \cancel{6} | \bar{e}_0]$$

$$= P(e_0) \left\{ P[X_1 = \cancel{6} | e_0] P[X_2 = \cancel{6} | e_0, X_1 = \cancel{6}] \right\} + P(\bar{e}_0) \left\{ P[X_1 = \cancel{6} | \bar{e}_0] P[X_2 = \cancel{6} | \bar{e}_0, X_1 = \cancel{6}] \right\}$$

$$= \frac{1}{2} \cdot \frac{1}{6} P[X_2 = \cancel{6} | \bar{e}_0] + \frac{1}{2} \cdot \frac{1}{2} P[X_1 = \cancel{6} | \bar{e}_0]$$

$$= \frac{1}{2} \cdot \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} \left[\frac{1}{36} + \frac{1}{4} \right] = \frac{1}{2} \cdot \frac{10}{36} = \frac{5}{36}$$

$$\frac{1}{2} \left[\frac{1}{36} + \frac{1}{20} \right] = \frac{1}{2} \left[\frac{5}{180} + \frac{9}{180} \right] = \frac{7}{180}$$

$$\frac{1}{2} \left[\frac{1}{60} + \frac{1}{12} \right] = \frac{1}{2} \frac{6}{60} = \frac{3}{60} = \frac{1}{20}$$